$\boxed{P7.2}$ Since $\hat{a}^+ \to \begin{pmatrix} 0 & 0 & 0 & \\ \sqrt{1} & 0 & 0 & \\ 0 & \sqrt{2} & 0 & \\ & & & \ddots \end{pmatrix}$

$$\hat{a} \to \begin{pmatrix} 0 & \sqrt{1} & 0 & \\ 0 & 0 & \sqrt{2} & \\ 0 & 0 & 0 & \\ & & & \ddots \end{pmatrix}$$

and
$$\begin{cases} \hat{x} = \sqrt{\dfrac{\hbar}{2m\omega}}\,(\hat{a} + \hat{a}^+) \\ \hat{p} = \sqrt{\dfrac{m\omega\hbar}{2}}\, i\,(\hat{a}^+ - \hat{a}) \end{cases}$$

we have

$$\hat{x} \to \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 \\ \sqrt{1} & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

$$\hat{p} \to \sqrt{\frac{m\omega\hbar}{2}}\, i \begin{pmatrix} 0 & -\sqrt{1} & 0 & 0 \\ \sqrt{1} & 0 & -\sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & -\sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

Thus
$$\hat{x}\hat{p} \to \frac{\hbar}{2} \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 \\ \sqrt{1} & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & -i\sqrt{1} & 0 & 0 \\ i\sqrt{1} & 0 & -i\sqrt{2} & 0 \\ 0 & i\sqrt{2} & 0 & -i\sqrt{3} \\ 0 & 0 & i\sqrt{3} & 0 \end{pmatrix} =$$

$$= \frac{\hbar}{2} \begin{pmatrix} i & 0 & -i\sqrt{2} & 0 \\ 0 & i & 0 & -i\sqrt{6} \\ i\sqrt{2} & 0 & i & \\ 0 & i\sqrt{6} & 0 & \ddots \end{pmatrix}$$

$$\hat{p}\hat{x} \to \frac{\hbar}{2} i \begin{pmatrix} 0 & -\sqrt{1} & 0 & 0 \\ \sqrt{1} & 0 & -\sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & -\sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 \\ \sqrt{1} & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} =$$

$$= \frac{\hbar}{2} i \begin{pmatrix} -1 & 0 & -\sqrt{2} & 0 \\ 0 & -1 & 0 & -\sqrt{6} \\ \sqrt{2} & 0 & -1 & 0 \\ 0 & \sqrt{6} & 0 & \ddots \end{pmatrix}$$

So that finally

$$[\hat{x},\hat{p}] \to \frac{\hbar}{2} \begin{pmatrix} i & 0 & -i\sqrt{2} & 0 \\ 0 & i & 0 & -i\sqrt{6} \\ i\sqrt{2} & 0 & i & 0 \\ 0 & i\sqrt{6} & 0 & \ddots \end{pmatrix} - \frac{\hbar}{2} \begin{pmatrix} i & 0 & -i\sqrt{2} & 0 \\ 0 & -i & 0 & -i\sqrt{6} \\ i\sqrt{2} & 0 & -i & 0 \\ 0 & i\sqrt{6} & 0 & \ddots \end{pmatrix}$$

$$= \hbar \begin{pmatrix} i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & i \end{pmatrix} = i\hbar \hat{1}$$

$\boxed{P7.4}$ $\hat{a}|0\rangle = 0 \Rightarrow \langle p|\hat{a}|0\rangle = 0$

$\boxed{\text{Derive } \langle p|0\rangle = \psi_0(p)}$

$$\Rightarrow \sqrt{\frac{m\omega}{2\hbar}} \langle p| \hat{x} + \frac{i}{m\omega}\hat{p} |0\rangle = 0$$

$$\Rightarrow i\hbar \frac{\partial}{\partial p} \langle p|0\rangle + \frac{i}{m\omega} p \langle p|0\rangle = 0$$

$\Rightarrow$ diff. equation for $\psi_0(p) \equiv \langle p|0\rangle$:

$$\frac{d}{dp}\psi_0(p) = -\frac{p}{m\omega\hbar}\psi_0(p)$$

$$\frac{d\psi_0(p)}{\psi_0(p)} = -\frac{1}{m\omega\hbar}\,p\,dp = -\frac{dp^2}{2m\omega\hbar}$$

$$\Rightarrow \psi_0(p) = C\,e^{-p^2/(2m\omega\hbar)}$$

Normalization is found by requiring

$$1 = \int_{-\infty}^{\infty} dp\,|\langle p|0\rangle|^2 = |C|^2\int_{-\infty}^{\infty} dp\; e^{-\frac{p^2}{m\omega\hbar}} =$$

$$= |C|^2\sqrt{m\omega\hbar}\;\underbrace{\int_{-\infty}^{\infty} dq\; e^{-q^2}}_{\sqrt{\pi}} = |C|^2\sqrt{\pi\hbar m\omega}$$

$$\Rightarrow C = (\pi\hbar m\omega)^{-1/4}$$

Thus $\langle p|0\rangle = \dfrac{1}{(\pi\hbar m\omega)^{1/4}}\,e^{-\frac{p^2}{2\hbar m\omega}}$ (Gaussian in momentum)

$\rightarrow$ compare with (7.44b): $\langle x|0\rangle = \left(\dfrac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$

$\boxed{\text{P7.10}}$ Recall Ch.2: matrix element of adjoint
(2.78) operator $\langle\psi|\hat{A}^+|\chi\rangle = \langle\chi|\hat{A}|\psi\rangle^*$

Hermitian = self-adjoint $\hat{A} = \hat{A}^+$

$\Rightarrow$ we need to show that

$$\langle\psi|\hat{A}|\chi\rangle = \langle\chi|\hat{A}|\psi\rangle^* \quad \text{for any } |\psi\rangle \text{ and } |\chi\rangle.$$

$$\langle \psi | \hat{\Pi} | \chi \rangle = \langle \psi | \hat{\Pi} \int dx \, |x\rangle\langle x|\chi\rangle = \left(\hat{\Pi}|x\rangle = |-x\rangle \right)$$

$$= \int dx \, \langle \psi | -x\rangle \langle x|\chi\rangle = \int dx \, \psi^*(-x)\, \chi(x)$$

On the other hand, make a change of variables $x = -x'$:

$$\langle \psi | \hat{\Pi} | \chi \rangle = \int_{-\infty}^{\infty} dx \, \langle \psi | -x\rangle \langle x|\chi\rangle = -\int_{\infty}^{-\infty} dx' \, \langle \psi | x'\rangle \langle -x'|\chi\rangle$$

$$= \int_{-\infty}^{\infty} dx' \, \langle \psi | x'\rangle \langle -x'|\chi\rangle = \int_{-\infty}^{\infty} dx \, \langle \psi | x\rangle \langle -x|\chi\rangle =$$

$$= \left(\int_{-\infty}^{\infty} dx \, \langle \chi | -x\rangle \langle x|\psi\rangle \right)^* = \langle \chi | \hat{\Pi} | \psi \rangle^*$$

$$\Rightarrow \quad \langle \psi | \hat{\Pi} | \chi \rangle = \langle \chi | \hat{\Pi} | \psi \rangle^* = \langle \psi | \hat{\Pi}^+ | \chi \rangle$$

Thus $\hat{\Pi} = \hat{\Pi}^+$.

$\boxed{\text{P7.11}}$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \langle x|E\rangle + \frac{1}{2} m\omega^2 x^2 \langle x|E\rangle = E \langle x|E\rangle \quad (1)$$

Try $\langle x|E\rangle = \psi_E(x) = N e^{-ax^2}$

$$\frac{d^2}{dx^2} e^{-ax^2} = \frac{d}{dx}\left(-2ax\, e^{-ax^2}\right) = 4a^2x^2 e^{-ax^2} - 2a e^{-ax^2}$$

$$\Rightarrow \quad (1): \quad -\frac{\hbar^2}{2m} N \left(4a^2x^2 - 2a\right) e^{-ax^2} + \frac{1}{2} m\omega^2 x^2 N e^{-ax^2} = E e^{-ax^2} N$$

Divide by $N e^{-ax^2}$:

$$-\frac{\hbar^2}{2m}\left(4a^2x^2 - 2a\right) + \frac{1}{2} m\omega^2 x^2 = E$$

Re-write: $\frac{x^2}{2}\left(m\omega^2 - \frac{4a^2\hbar^2}{m}\right) = E - \frac{\hbar^2 a}{m}$

Demand the coefficient of x^2 to be zero:

$$\Rightarrow \quad m\omega^2 = \frac{4a^2\hbar^2}{m} \Rightarrow a^2 = \frac{m^2\omega^2}{4\hbar^2} \Rightarrow a = \pm \frac{m\omega}{2\hbar}$$

This leads to $E - \frac{\hbar^2 a}{m} = 0$

$$\Rightarrow \quad E = \frac{\hbar^2}{m}\left(\pm \frac{m\omega}{2\hbar}\right) = \pm \frac{m\omega\hbar}{2} = \pm \frac{1}{2}\hbar\omega.$$

And we need the $\oplus$ sign (because the energy eigenvalue can not be smaller than the minimum of potential $V(x) = \frac{1}{2}m\omega^2 x^2$)

$$\Rightarrow \quad E = \frac{1}{2}\hbar\omega$$ = the lowest energy state of harmonic oscillator.

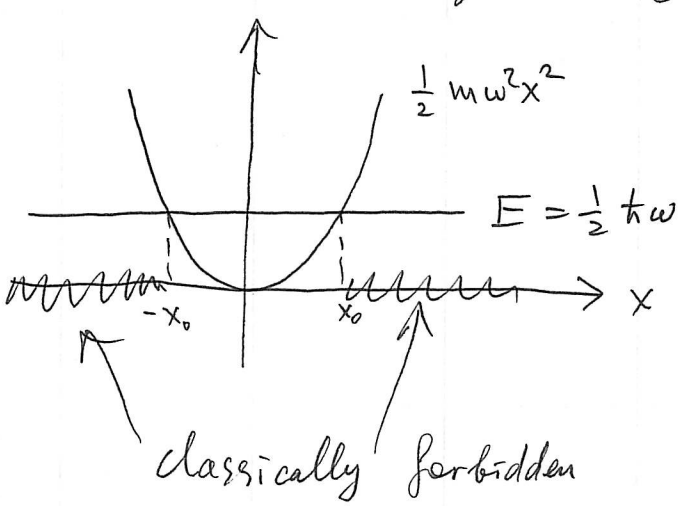
$$a = \frac{m\omega}{2\hbar}$$

$$\psi_E(x) = N e^{-\frac{m\omega}{2\hbar}x^2}.$$

$\boxed{P7.12}$

$$\langle x|0\rangle = \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}}$$

Its energy $E = \frac{1}{2}\hbar\omega$



Determine x_0:

$$\frac{1}{2}\hbar\omega = \frac{1}{2}m\omega^2 x_0^2$$

$$\Rightarrow x_0^2 = \frac{\hbar\omega}{m\omega^2} = \frac{\hbar}{m\omega}$$

$$x_0 = \sqrt{\frac{\hbar}{m\omega}}$$

Probability to find particle somewhere in the region $|x| > x_0$ is

$$P_> = \int_{-\infty}^{-x_0} dx\, |\langle x|0\rangle|^2 + \int_{x_0}^{\infty} dx\, |\langle x|0\rangle|^2 =$$

$$= 2\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \int_{x_0=\sqrt{\frac{\hbar}{m\omega}}}^{\infty} dx\, e^{-\frac{m\omega x^2}{\hbar}} =$$

$$= \left(x = \sqrt{\frac{\hbar}{m\omega}}\, y \;;\; dx = \sqrt{\frac{\hbar}{m\omega}}\, dy\right) =$$

$$= 2\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \sqrt{\frac{\hbar}{m\omega}} \int_1^{\infty} dy\, e^{-y^2} = \frac{2}{\sqrt{\pi}} \int_1^{\infty} dy\, e^{-y^2}$$

$$= \frac{2}{\sqrt{\pi}} \left(\underbrace{\int_0^{\infty} dy\, e^{-y^2}}_{\frac{\sqrt{\pi}}{2}} - \underbrace{\int_0^{1} dy\, e^{-y^2}}_{I} \right) = 1 - \frac{2}{\sqrt{\pi}} I$$

$I = \int_0^1 dy\, e^{-y^2}$ can be looked in the Table of integrals.

I will get it using Mathematica:

$$I = \int_0^1 dy\, e^{-y^2} = \frac{\sqrt{\pi}}{2}\, \mathrm{Erf}(1) = 0.7468$$

↳ the error function

$$\Rightarrow P_> = 1 - \frac{2}{\sqrt{\pi}}(0.7468) = 0.1573 \approx 0.16$$

$\Rightarrow$ the probability to be in the forbidden region is about 16%.